

Interactive 2D Convolution Playground

CSE 219: Signals and Linear Systems | Revised Proposal

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Description

An interactive playground for exploring 2D convolution. Users supply an input matrix, apply preset or custom kernels, and watch the convolution computed step by step. Beyond visualizing the sliding multiply-and-add, the tool treats the kernel as the impulse response of a 2D LTI system and lets users test the reflect-and-shift definition, causality, and BIBO stability.

Input

- Sketchable 2D grid, or image upload converted to a pixel-intensity matrix.
- Preset kernel library: edge detection, blur, sharpen, emboss.
- Custom kernel editor for user-defined values and sizes.

Computation and Visualization

- Adjustable stride, padding mode, and kernel size.
- Step-by-step animation of the kernel sliding across the input.
- Side-by-side view of input, kernel, and output matrix.
- Heatmap / grayscale rendering of the convolved result for image inputs.

Additional Features

[NEW] Reflect-and-Shift Stage with Correlation Comparison

- Animates the explicit flip $k[i,j] \rightarrow k[-i,-j]$ before the shift and multiply-add.
- Toggle between true convolution and cross-correlation.
- Reports the numeric difference between the two — zero for symmetric kernels, nonzero otherwise.

[NEW] Causality and BIBO Stability Analyzer

- Tests whether the kernel is causal under raster-scan ordering.
- Computes the minimum origin shift (a, b) needed to make a non-causal kernel causal.
- Computes $S = \sum \sum |h[i,j]|$ and compares the bound $|y| \leq B \cdot S$ against the actual output maximum.

[NEW] Impulse and Step Response Mode

- Convolution with a 2D unit impulse returns the kernel itself, showing kernel = impulse response.
- Convolution with a 2D unit step shows the accumulated response.
- Displays DC gain $\sum \sum h[i,j]$, classifying the kernel as averaging or differencing.